# Topic: Gradient

## I. Scalar Functions

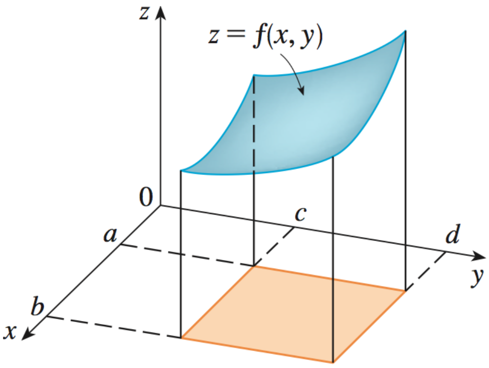
Defn: A Scalar Function is a function

Ex: $f(x) = x^2$ is a scalar function since

$x \in$ and $f(x) \in$

Ex: $f(x,y) = x - x\sqrt{y}$ is a scalar function since

$(x,y) \in$ and $f(x,y) \in$




Ex: $f(x,y,z) = \frac{\cos(z)}{x} - x$ is a scalar function since

$(x,y,z) \in$ and $f(x,y,z) \in$

## II. Gradient

Defn: Let $D_f$ be the domain of the scalar function $f(x,y)$ and $D_g$ be the domain of the scalar function $g(x,y,z)$. If $f$ is differentiable on $D_f$ and $g$ is differentiable on $D_g$ then the **gradient** of $f(x,y)$ and $g(x,y,z)$ is given by

$$\nabla f =$$

$$\nabla g =$$

Ex: $f(x,y) = xy + y$. Find $\nabla f$

Ex: $f(x,y,z) = z\ln(xy^2 + y)$. Find $\nabla f$ at the point $P(0,1,2)$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right]$

1. $\nabla(f+g) = \nabla f + \nabla g$

   *Proof* : $\nabla(f+g) =$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

   *Proof* : $\nabla(fg) =$

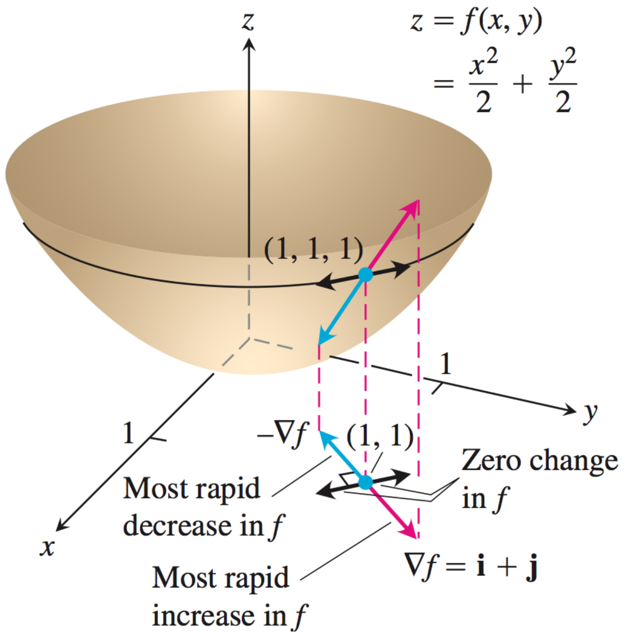
# III. Additional Concepts

| Thm: $\nabla f(P)$ points in the direction of |
|---|

| Defn: The **Directional Derivative** of a function $f$, $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is |
|---|
| $D_{\mathbf{u}}f =$ so we also have $D_{\mathbf{u}}f =$ |

Ex: Let $f(x,y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1,1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right]$

Compute $D_{\mathbf{u}}f(P)$



Ex: Is there a direction **u** in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14°C/ft$? Give reasons for your answer.

ANSWER:

REASON: